

to Microsoft Office, achieving 14 per cent penetration in the large enterprise market by 2004. The development of OpenOffice.org was sponsored primarily by Sun Microsystems, which used the code as the basis for subsequent versions of StarOffice.

After acquiring Sun in January 2010, Oracle Corporation continued developing OpenOffice.org and StarOffice, which it renamed Oracle Open Office. In June 2011, Oracle contributed the trademarks of this software to the Apache Software Foundation. It also contributed Oracle-owned code to Apache for relicensing under the Apache License, at the suggestion of IBM, as the latter did not want the code put under a copyleft licence. This code drop formed the basis for the Apache OpenOffice project.

OpenOffice.org (OOo), commonly known as OpenOffice, includes a word processor (Writer), a spreadsheet (Calc), a presentation application (Impress), a drawing application (Draw), a formula editor (Math) and a database management application (Base). Its default file format is the Open Document Format (ODF), an ISO/IEC standard, which originated with OpenOffice.org. It can also read a wide variety of other file formats, particularly those from Microsoft Office. Apache renamed the software Apache OpenOffice.

OpenOffice.org was primarily developed for Linux, Microsoft Windows and Solaris, and later for OS X, with ports to other operating systems. It is distributed under the GNU Lesser General Public License version 3 (LGPL).

Table 1: The components of OpenOffice

Name of component	Description
Writer	Word processor analogous to Microsoft Word or Word Perfect
Calc	Spreadsheet analogous to Microsoft Excel
Impress	Presentation program analogous to Microsoft PowerPoint
Draw	Vector graphics editor
Math	Tool for writing mathematical formula and equations, analogous to Microsoft Equation Editor
Base	Database management program analogous to Microsoft Access

Latest version: 4.1.7

Website: www.openoffice.org

2. LibreOffice

On September 28, 2010, it was announced that The Document Foundation would be the host of LibreOffice, a new derivative of OpenOffice.org. The Foundation's initial announcement echoed users' concerns that Oracle would either discontinue OpenOffice.org, or place restrictions on its use as an open source project, as it had done with Sun's OpenSolaris. The alternative name LibreOffice was picked after researching trademark databases and social media, as well as after checking to see if it could be used for URLs in other countries too. Earlier, OpenOffice.org had been distributed as *BrOffice.org* in Brazil by the BrOffice Centre of Excellence for Free Software because of a trademark issue. LibreOffice 3.3 beta used the ooo-build infrastructure and the *OpenOffice.org* 3.3 beta code from Oracle, then adding selected patches from Go-oo. Later, Go-oo was discontinued in favour of LibreOffice.

Since the office suite that was branded OpenOffice.org in most Linux distributions was in fact Go-oo, most users moved immediately to LibreOffice. It was forked in 2010 from OpenOffice.org, and uses the international ISO/IEC standard Open Document File format (ODF) to save documents for all of its applications. LibreOffice also supports the file formats of most other major office suites, including Microsoft Office, through a variety of import/export filters.

LibreOffice is available for a variety of computing platforms, including Microsoft Windows, macOS, and Linux (including a LibreOffice Viewer for Android). The LibreOffice project uses a dual LGPLv3 (or later) / MPL 2.0 licence for new contributions in order to allow the licence to be upgraded. Since the core of the OpenOffice.org codebase was donated to the Apache Software Foundation, there is an ongoing effort to get all the code rebased to ease future licence updates.

LibreOffice supports third-party extensions. It lists more than 320 extensions and scripts for LibreOffice, which can be written in C++, Java, CLI, Python, and LibreOffice Basic. LibreOffice Basic is a programming language similar to Microsoft Visual Basic for Applications (VBA) and is available in Writer, Calc and Base. It is used to write small programs known as macros, with each macro performing a different task, such as counting the words in a paragraph.

LibreOffice has the same six applications as Apache OpenOffice—Writer, Calc, Impress, Draw, Base and Math—but the two suites differ in terms of the applications' features, usability and community support. Specifically, LibreOffice is often considered to be more actively developed than OpenOffice; the standard package, for example, offers PDF import, a presentation minimiser and a Wiki publisher. Users can expand functionality with extensions and templates as well.